Direct and Anchor-Gated Polyp Sizing from Monocular Colonoscopy Video

Literature Review and Research Direction

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Abstract

Polyp size is clinically consequential, but estimating it from a monocular colonoscopy video is not the same task as estimating relative depth. This review introduces the clinical and camera context needed by machine-learning researchers, surveys direct and reference-based size-estimation approaches, and identifies their shared limitations. Wide-angle optics, near-light illumination, non-rigid tissue, quality variation, and missing metric scale make single-frame pixel measurements unreliable. Existing approaches use learned RGB/depth features, temporal aggregation, calibrated optical priors, or visible references such as a waterjet or snare. The recommended research program is an evaluation-first, short-video direct-sizing model: track a lesion, filter for usable views, predict an ordinal or continuous endpoint, aggregate at polyp level, and emit millimetres only when a validated scale anchor is available. In all other cases, the model should return a calibrated ordinal estimate or abstain rather than imply unsupported metric precision.

1 Scope and terminology

This is a methods review for research planning, not clinical advice. A *polyp* is a lesion on the inner lining of the large bowel. During colonoscopy, an endoscopist inspects the lesion, may remove it, and records an estimated diameter. In research, ≤ 5 mm, 6–9 mm, and ≥ 10 mm are useful ordinal ranges; they should not be treated as a substitute for clinical guidelines or an automatic treatment policy.

The terms *metric size*, *relative depth*, and *ordinal size* must remain separate. Metric size is a physical dimension in millimetres. Relative depth is a geometrically consistent shape up to a global scale. Ordinal size predicts a clinically useful range without asserting millimetres. This review distinguishes methods that output a number from methods that have validated a transferable scale channel.

2 Clinical and camera context

2.1 What is measured in practice

Labels called `size_mm` are not interchangeable. A fresh-specimen caliper records an excised object that may have changed through handling; pathology records can reflect fixation or fragmentation;

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an endoscopist's visual estimate is clinically useful but not independent metric ground truth; and a snare or forceps ratio depends on tool opening, visibility, pose, and whether the tool lies at the lesion depth. Phantom geometry offers controlled ground truth but does not reproduce clinical tissue or workflow. Every evaluation manifest should record label provenance, measurement protocol, scope model, site, operator, and whether an in-scene reference is present.

2.2 Why a colonoscopy image is difficult

A colonoscope is a wide-angle endoscope with illumination close to the lens. Its imagery differs from a fixed, well-calibrated pinhole camera in several ways: field-of-view distortion and image processing vary by scope; brightness changes with distance, orientation, exposure, white balance, and tissue reflectance; tissue deforms under insufflation, contact, fluid, peristalsis, and scope pressure; and procedure video includes blur, bubbles, debris, instruments, treatment, and occlusion. A lesion clip is therefore a selected subset of an uneven, quality-variable procedure video, not simply many independent still images.

2.3 Metric-scale ambiguity

For a perspective camera, $x = fX/Z$. Multiplying X, Y, Z, and camera translation by the same positive constant leaves image projection unchanged. Monocular structure-from-motion and photometric self-supervision can recover relative geometry without recovering its physical scale. Median scaling can evaluate relative shape, but never proves millimetres. Metric output needs an independent scale cue—for example a known visible reference, a calibrated lens-and-procedure prior that is validated for the target setting, or a scope-specific empirical calibration.

3 Methods in the literature

Table 1 groups representative approaches by the evidence that links visual features to size. The table is deliberately not a leaderboard: cohorts, label protocols, endpoint definitions, and split units differ too substantially for the reported numbers to be ranked across papers.

Early direct approaches demonstrate why a reconstruction-only framing is too narrow. BseNet predicts self-supervised relative depth, combines it with RGB, and pools temporal information for a binary size decision [1]. This makes the output closer to a clinical question, but does not resolve metric identifiability. More recent direct models use segmentation, estimated depth, and learned fusion to regress a diameter or class [7, 2]. Their central contribution is the endpoint formulation; their measurement claims still depend on the label and scale protocol.

Reference-based work makes that protocol explicit. ENDOANGEL uses calibrated lens parameters in a multicentre prospective validation [6]. Poseidon uses the waterjet as a physical in-frame reference [5]. These strategies carry different operational costs, but both expose a measurement channel that a generic unanchored video cannot guarantee.

Temporal representation learning is also relevant. Parolari et al. study contrastive learning under noisy temporal self-supervision and evaluate downstream colonoscopy-video tasks including size prediction [3]; this is a 2026 preprint rather than clinical validation. Its useful lesson for a sizing project is to model a lesion track, not merely to average arbitrary adjacent frames.

Approach and representative work	Endpoint, scale evidence, and planning implication
RGB plus self-supervised depth and temporal pooling: Itoh et al. (BseNet) [1]	Binary < 10 versus ≥ 10 mm classification; depth is an auxiliary learned cue rather than an explicit physical reference. Direct clinical endpoint and temporal aggregation are feasible, but binary accuracy does not validate metric scale or cross-scope transfer.
Attention-gated multi-cue classifiers: Roffo et al. [4] (preprint)	RGB, depth, and location features for binary and three-class sizing. Combining cues is plausible; evaluate at lesion level and enforce site/scope/entity-disjoint splits before drawing general conclusions.
Reference-free direct regression: Yii et al. [7]	Segmentation and depth cues with direct millimetre regression, evaluated under a single-scope, snare-anchored protocol. A direct regression head is viable, but the protocol's anchor and hardware context are part of the result.
Geometry-aware direct regression: Lawrence et al. (MPSE) [2]	Per-frame RGB/mask/pseudodepth regression with polyp-level aggregation. Aggregation should be designed around a tracked lesion rather than chosen post hoc; small or imbalanced cohorts require uncertainty reporting.
Calibrated optical prior: ENDOANGEL [6]	A real-time deep-learning system that incorporates a lens-parameter calibration channel. Treat calibration as a target-environment requirement and validate it prospectively across hardware and sites.
Visible physical reference: Poseidon [5]	The auxiliary waterjet serves as a reference object for size measurement. A deliberately visible anchor is often the clearest path from pixels to millimetres, but introduces workflow and visibility failure modes.

Table 1: Representative methods and the scale evidence they use. Preprint status is stated explicitly; its findings should not be treated as mature clinical evidence.

4 Failure modes that should shape evaluation

The most common risk is false precision: a model returns millimetres because its head is trained to regress a number, while its evaluation does not prove that scale transfers across scopes, sites, or label sources. Other predictable failure modes are frame-level leakage from the same lesion into train and test; mixed label provenance; tool-anchor occlusion or orientation error; blurred or near-contact views; under-representation of large polyps; and cherry-picking the best frame after observing labels. A dataset split should be at least polyp- and procedure-disjoint, and when data permit, site-, scope-, and operator-aware. Report performance by clinically relevant range, label source, anchor visibility, scope model, and image-quality stratum.

For a continuous metric endpoint, report MAE together with agreement analysis, calibration, confidence intervals at the lesion level, and an abstention curve. For an ordinal endpoint, report confusion across clinically defined ranges, calibration of class probabilities, and selective risk as low-confidence cases are deferred. A fixed qualitative panel should include specular highlights, blur, bubbles, instruments, partial lesions, low insufflation, and difficult viewing angles. Do not use median-scaled depth as evidence of metric accuracy.

5 Recommended research program

The recommended path is an *anchor-gated short-video direct-sizing* track, run beside the project's existing geometry guardrails rather than assuming the depth-pose stack is the sole solution.

1. **Create an evaluation manifest first.** Deduplicate at lesion and procedure level; preserve label provenance; mark scope/site/operator; record anchor type and visibility; and reserve leakage-checked held-out lesions before selecting an architecture.

2. **Form lesion clips.** Detect or segment the lesion, track it through a short clip, and reject frames with blur, occlusion, poor lesion coverage, excessive specularity, or unreliable tool/anchor visibility.

3. **Use two linked outputs.** Train an ordinal head for all quality-approved clips. Train a continuous size head only for examples where the scale evidence is valid; predict uncertainty for both outputs.

4. **Aggregate at polyp level.** Combine frame predictions using a pre-specified robust rule, attention, or learned temporal model. The unit of selection and reporting should be the lesion, not the most favorable frame.

5. **Gate metric output.** Emit millimetres only when anchor detection, calibration, image quality, and uncertainty all pass. Otherwise output the ordinal distribution or abstain.

6. **Compare against useful baselines.** Include a single-frame RGB ordinal model, a clip model without depth features, an optional depth-feature model, and a reference-based baseline when anchors are available.

This approach is deliberately conservative. It permits a model to be useful in the many views where millimetres are not defensible, while making the conditions for a metric claim inspectable. The project's self-supervised depth model can enter as a feature or a quality signal, but it should be evaluated as an incremental contribution rather than assumed to solve scale.

6 Conclusion

Direct polyp sizing from video is a legitimate endpoint, but direct regression alone does not solve monocular scale ambiguity. The strongest next experiment is a leakage-controlled, polyp-level benchmark with explicit anchor and label provenance, followed by a short-video model that separates ordinal inference from anchor-supported metric output. Such a design is more clinically legible, more robust to missing scale evidence, and easier to falsify than a claim that unanchored monocular reconstruction recovers millimetres.

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